

2026 ENVR Workshop

Invited sessions and talk abstracts

Gaussian Processes and Deep Learning

Chris Geoga

University of Wisconsin-Madison, Department of Statistics

High-resolution retrieval of atmospheric boundary layers with nonstationary Gaussian processes

Abstract. The atmospheric boundary layer (ABL) plays an important role in governing many meteorological processes in the troposphere. High resolution retrieval of the ABL height is difficult, however, as oftentimes it must be inferred from one-dimensional wind profiles. This work introduces a Gaussian process-based approach for exploiting spatio-temporal dependence in wind profiles to identify fine-scale fluctuations in the ABL height, dramatically improving the temporal resolution over current estimation methods from tens of minutes down to tens of seconds. Computational methods to make this method feasible for full-day profiles, which can often provide over 5M measurements, will also be discussed and demonstrated.

Michele Peruzzi

University of Michigan, Ann Arbor, Department of Biostatistics

Inside-out models for multivariate geostatistics

Abstract. Environmental statistics increasingly requires multivariate spatial models that can represent dependent processes over large domains, including air pollutants, climate variables, species abundances, land-use features, and remotely sensed indicators. These data exhibit dependence across space, across outcomes, and through covariates, but standard models such as the linear model of coregionalization (LMC) often force a trade-off between interpretability, flexibility, and scalability. In this talk, I will introduce novel cross-covariance models for multivariate spatial data based on the inside-out construction, whereby a valid cross-covariance function arises by inducing outcome-specific spatial dependence on independent replicates of a correlated random vector. Relative to the LMC, this enables more direct marginal inference, transparent prior elicitation, unequal smoothness across outcomes, novel ways to perform dimension reduction. The same construction also supports a process-level graphical perspective the LMC cannot deliver: inside-out models for Gaussian processes straightforwardly characterize process-level partial correlations and easy-to-enforce conditions for conditional independence across margins of the multivariate process. These results give novel methodological support to environmental questions in which direct relationships among processes (e.g., interactions between species) are more meaningful than marginal cross-correlation functions. Finally, inside-out models facilitate the development of novel massively scalable computational algorithms, making inference practical when both the number of locations and the number of outcomes are large. In particular, we will highlight scalable posterior inference via variational approximations and block Markov-chain Monte Carlo samplers as implemented in the R package `spiox`, available at github.com/mkln/spiox.

Gaussian Processes and Deep Learning continued

Mikael Kuusela

Carnegie Mellon University

Neural Likelihood Inference for Large-Scale Spatial Data

Abstract. Neural simulation-based inference is a recent paradigm shift in statistical science that uses simulations from a statistical model to train a neural network for statistical inference. These neural models serve as fast and accurate surrogates for classical inferential quantities, such as point estimators, confidence sets, likelihoods, posteriors or conditional distributions. In this talk, I will describe our work on neural likelihood inference for scalable parameter estimation in spatial modeling. I will show how to train an amortized neural surrogate model for the likelihood function of the spatial process and demonstrate that this leads to substantial speed-ups for Gaussian processes and accuracy gains for non-Gaussian processes. I will then demonstrate the benefits of this approach in complex real-world settings using large-scale oceanographic observations. Specifically, I will describe how neural likelihood inference can be adapted for scalable spatio-temporal modeling of remotely sensed sea surface temperature data and subsurface Argo profiling float data. I will conclude by describing recent methodological improvements that increase training sample efficiency by augmenting the loss function with score information.

Noel Cressie

University of Wollongong, Australia

Spatio-temporal modeling and forecasting with Fourier neural operators (FNOs)

Abstract. Spatio-temporal process models are often used for modelling dynamic physical and biological phenomena that evolve across space and time. These phenomena may exhibit environmental heterogeneity and complex interactions that are difficult to capture using traditional statistical process models such as Gaussian processes. This work proposes the use of Fourier neural operators (FNOs) for constructing statistical dynamical spatio-temporal models for forecasting. An FNO is a flexible mapping of functions that approximates the solution operator of possibly unknown linear or non-linear partial differential equations (PDEs) in a computationally efficient manner. It does so by training on samples of inputs and their respective outputs, and hence explicit knowledge of the underlying PDE is not required. Through simulations from a nonlinear PDE with a known solution, we compare FNO forecasts to those from state-of-the-art statistical spatio-temporal-forecasting methods. Further, using precipitation data in central Europe, we demonstrate the ability of FNO-based dynamic spatio-temporal (DST) statistical modelling to capture complex real-world spatio-temporal dependencies. The research presented in this talk is joint with Dr Pratik Nag (University of Wollongong, Australia), Associate Professor Andrew Zammit Mangion (University of New South Wales, Australia), and Professor Sumeetpal Singh (University of Wollongong, Australia)

Advances in Point Process Modeling

Erin Schliep

North Carolina State University

Call me maybe: Hawkes processes with excitation and inhibition

Abstract. When studying animal communication processes, some calls induce excitement, e.g., lead to increased calling activity from the individual or neighboring individuals. Other calls may inhibit future calling activity. With only excitement, Hawkes process models are well established in the literature. Models incorporating both excitement and inhibition are much less common. We develop a multivariate version of the Hawkes process model which incorporates additive excitement and multiplicative inhibition. We investigate the model through various simulations and then apply it to study the communication behavior in two different settings -- meerkats and whales.

Nicholas Clark

University of St. Thomas

Hawkes Trajectories - A Framework for Leadership Inference in Coordinated Movement

Abstract. Coordinated movement among multiple agents is naturally modeled as a system of interacting event processes, where movement by one individual increases the likelihood of movement by others shortly thereafter. Continuous-time Hawkes processes provide a principled framework for such interactions. We introduce Hawkes trajectories, a multivariate marked Hawkes framework in which movement events are indexed in time and enriched with kinematic marks including spatial location, speed, and heading. We demonstrate how leadership and followership scores can be constructed from the estimated excitation network acknowledging that lead-lag does not, by itself, establish causal leadership. To accommodate large-scale trajectory data and enable more flexible modeling, we propose a discrete-time approximation based on Poisson autoregressive models with cross-lag dependence and mark-similarity weighting. This formulation preserves the interpretability of Hawkes excitation while permitting efficient estimation and diagnostics using standard GLM tools.

Advances in Point Process Modeling continued

Joe Eisaguirre

Alaska Science Center, USGS

Modeling spatial variation in animal mortality risk using time-marginalized and thinned spatial point processes

Abstract. Mortality risk for animals often varies spatially and can be linked to how animals use landscapes. While numerous studies collect telemetry data on animals, the focus is typically on the period when animals are alive, even though there is important information that could be gleaned about survival. Formally linking animal space use and mortality processes in a single statistical model could provide new insights about animal ecology and species management. In this talk, I will introduce a thinned spatial point process modelling framework that couples relative abundance and space use with a mortality process to formally treat the occurrence of mortality events across the landscape as a spatial process. In doing so, I will also review a time-marginalization property of space-time point processes that adds flexibility to the approach. I will then show how this model can be embedded in a hierarchical statistical framework and fit to telemetry data to make inferences about how spatial covariates drive both space use and mortality risk. Finally, I will discuss applications of the method to two data sets to study the effects of roads and habitat on spatially explicit mortality risk: (1) VHF telemetry data collected for willow ptarmigan in Alaska, and (2) hourly GPS telemetry data collected for black bears in Colorado. These case studies demonstrate the applicability of this method for different species and data types, making it broadly useful in enabling inferences about the mechanisms influencing animal survival and spatial population processes while formally treating survival as a spatial process.

Dani Gamerman

Universidade Federal do Rio de Janeiro

A Bayesian Joint Model For Multiple Point Processes With Application To Presence-Only Data

Abstract. Joint modeling of multiple point processes is relevant in applications where relationships among processes are of interest. Statistical inference becomes particularly challenging when multiple processes are analyzed jointly and the observed data correspond to presence-only patterns, which are subject to preferential sampling and partial observability. This paper proposes a Bayesian joint model for multiple point processes, with application to the presence-only setting. Dependence structures are explicitly incorporated into the probabilistic specification of the model using Bayesian networks. Latent data processes are introduced, allowing the augmented likelihood function to be exactly evaluated. This formulation also enables direct inference on the number and spatial distribution of unobserved occurrences. Inference is carried out using Markov chain Monte Carlo with blocked Gibbs sampling. Simulation studies demonstrate that the proposed inferential scheme is able to recover the true model parameters satisfactorily. The proposed model is applied to real presence-only data from Amazonia, involving archaeological indicators of pre-Columbian land management (earthworks and Amazonian dark earths) and occurrences of tree species. The results are consistent with findings reported in the literature, illustrating how the proposed model enables inference on the magnitude of relationships between processes, beyond the association with environmental covariates. This talk is based on joint work with Laura C. M. Teixeira, Carolina Levis and Vinicius Peripato.

Spatial-temporal Statistics for Public Health

Dan Kowal

Department of Statistics & Data Science, Cornell University

Efficient Bayesian inference for two-stage models in environmental epidemiology

Abstract. Statistical models often require inputs that are not completely known. This can occur when inputs are measured with error, indirectly, or when they are predicted using another model. In environmental epidemiology, air pollution exposure is a key determinant of health, yet typically must be estimated for each observational unit by a complex model. Bayesian two-stage models combine this stage-one model with a stage-two model for the health outcome given the exposure. However, analysts usually only have access to the stage-one model output without all of its specifications or input data, making joint Bayesian inference apparently intractable. We show that two prominent workarounds—using a point estimate or using the posterior from the stage-one model without feedback from the stage-two model—lead to miscalibrated inference. Instead, we propose efficient algorithms to facilitate joint Bayesian inference and provide more accurate estimates and well-calibrated uncertainties. Comparing different approaches, we investigate the association between PM_{2.5} exposure and county-level mortality rates in the South-Central USA.

Charles Doss

School of Statistics, University of Minnesota

Spatio-temporal state-space compartment models for infectious disease surge detection from wastewater data

Abstract. SIR-type (Susceptible-Infectious-Recovered-type) compartmental models are commonly used in infectious disease modeling. State-space models are appropriate when underlying variables of interest are only noisily or indirectly observed. This is the setting with wastewater infectious disease surveillance measurements, which are a noisy measurement of disease prevalence. We develop a spatio-temporal state-space compartmental model for use with wastewater data, especially for the purpose of quickly detecting infection surges, which is a core goal in wastewater surveillance monitoring because the wastewater signal often precedes clinical case counts. The model is nonlinear but can be effectively fit with a blocked sequential Monte Carlo approach with guided proposals which allows for moderate or high dimensionality. We demonstrate the approach on data from the COVID-19 pandemic.

Spatial-temporal Statistics for Public Health continued

Katherine Ensor

Department of Statistics, Rice University

Wastewater Epidemiology in Support of Public Health

Abstract. Wastewater epidemiology has emerged as an important public health tool in understanding disease transmission in an urban environment and planning for needed public health services. The critical information is the copies per liter of RNA signal derived from wastewater. From here we scale up to areal spatial-temporal real-time mapping and forecasting of disease moving fully to prevalence in the population. In this talk we will highlight the unique methodologies we have brought forward in population weighted extended Hausdorff-distance for spatial weights coupled with state-space methodologies to capture the strong temporal dependence. This is joint work with Jose Palacio, Julia Schedler, Cesar Cardenas, Lauren Stadler and her lab, and data analysts and public health officials at Houston Health Department. Important considerations for sustainability of the longstanding Rice-City of Houston partnership will also be discussed.

Cici Bauer

Department of Biostatistics & Data Science, University of Texas Health Science Center at Houston

Beyond the Model: From Complex Wastewater Data to Actionable Public Health Intelligence

Abstract. Wastewater-based epidemiology (WBE) has rapidly evolved from an emerging surveillance approach into an important component of the public health data ecosystem, creating new opportunities as well as distinct biostatistical challenges. Wastewater measurements are complex, longitudinal, geographically structured, and influenced by sampling design, laboratory methods, normalization strategies, population dynamics, limits of detection, and other sources of variability. As WBE expands across pathogens and public health applications, rigorous statistical approaches are needed not only to characterize trends and quantify uncertainty, but also to integrate wastewater signals with clinical, epidemiologic, and other surveillance data and translate these analyses into meaningful public health information. Equally important, and sometimes underemphasized in biostatistical practice, is how model outputs and their uncertainty are communicated to different audiences. Effective translation requires moving beyond statistical performance alone to consider what information is actionable, how uncertainty and changing trends should be represented, and how communication should differ for public health agencies, healthcare professionals, and the public. Drawing on applied WBE experience, I will discuss evolving analytical challenges and highlight the role of biostatisticians across the full data-to-decision continuum, from modeling complex surveillance data to designing dashboards, visualizations, and other data-driven communication tools that support timely and interpretable public health decision-making.

Advances in Environmental Statistics

Steve Sain

Jupiter Intelligence

Advances in climate analytics for resilience and risk management

Abstract. Climate risk analytics draws on climate science, statistical modeling, and financial/operational risk assessment to quantify how acute weather events and long-term climate change affect organizations, and to translate that understanding into actionable decisions. This talk walks through the analytics pipeline from hazard modeling and downscaling, through exposure and impact assessment, to adaptation evaluation and includes a focus on where applied statistics and machine learning drive methodological advances at each stage. I'll highlight specific applications in tropical cyclone risk, extreme heat, and wildfire, and where AI-enabled climate products are beginning to play a key role.

Zhengyuan Zhu

Iowa State University

Spatial prediction of local soil erosion distributions in the Wasserstein space

Abstract. Accurate measurement of soil erosion requires intensive and costly fieldwork, making it impractical to directly survey large spatial domains such as provinces or river basins. To extend localized field measurements to broader regions, we propose a novel spatial prediction framework that treats local erosion distributions as random objects in the Wasserstein space. These distributions are mapped to square-integrable trajectories and represented through basis expansion, yielding a multivariate random field that captures spatial dependence. By performing local regression and kriging within this representation, the proposed approach flexibly models and predicts erosion distributions at arbitrary spatial locations. This framework enables improved prediction for functionals of the distribution, such as the mean and exceedance probabilities. Simulation studies demonstrate that the proposed method outperforms mis-specified parametric models and existing Fréchet regression approaches. We also compare it with nonparametric prediction using TabPFN, highlighting the respective strengths and limitations of each method. We illustrate the methodology with a comprehensive erosion analysis in Shaanxi Province, China, where local watershed measurements are used to predict erosion distributions across the entire province, incorporating covariates such as land use and elevation.

Advances in Environmental Statistics continued

Veronica Berrocal

University of California, Irvine

A Bayesian scalar-on-function kernel regression model for crop yield

Abstract. Agriculture in the United States is critical for food security across the world. Additionally, the agricultural industry contributes substantially to the United States' GDP. Any deterioration in the U.S. agricultural yield could have considerable consequences for worldwide economy and food security. To better understand climate's impact on crop yield and devise successful climate adaptation strategies, it is fundamental to know at what stages of a plant's growth crops are most vulnerable to the harmful effects of certain environmental factors. In this talk, we present a modeling strategy to determine how two important climatic conditions, heat stress and water stress, individually and jointly affect crop's health during the course of a growing season. Our model is specified within a statistical functional framework, estimated under the Bayesian paradigm, and postulates a non-linear interaction between heat and water stress. Additionally, the model envisions the effect of heat and water stress to be received by the crops differently depending on the plant's developmental stage. We illustrate the capability of the model by applying it to maize yield in the US Midwest over a 4-year period.

Raj Guhaniyogi

Texas A&M University

When Spatial Statistics Meets Deep Learning: A Marriage Worth Making?

Abstract. Environmental and climate datasets are growing unprecedentedly large, richly structured, and stubbornly complex, demanding statistical methods that are simultaneously scalable and uncertainty-aware. Classical geostatistical approaches, grounded in Gaussian processes (their computationally efficient versions) and Bayesian hierarchical models, offer elegant uncertainty quantification (UQ), typically relying on computationally burdensome Markov Chain Monte Carlo (MCMC) sampling. This talk asks a question: can deep learning and spatial statistics be productively married? We present two MCMC-free frameworks for uncertainty-aware modeling of large spatially correlated data. The first introduces a calibrated geospatial hierarchical variational autoencoder that delivers scalable UQ for prediction and inference in spatially-varying coefficient models across massive datasets. The second proposes a calibrated Monte Carlo Dropout-based framework blending principled statistical modeling with deep neural networks for characterizing spatially varying cross-correlations among multivariate spatial outcomes at massive scale. Both approaches are benchmarked against classical stochastic process-based methods and existing spatial deep learning approaches, with an appraisal of where these approaches earn their place, where they fall short, and where the most promising research frontiers lie.